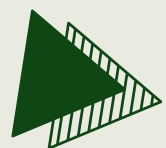


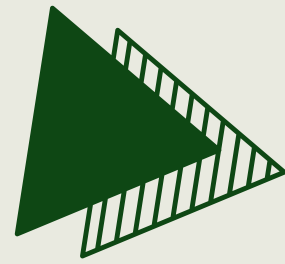
MED EX ANALYZER



Insights into Patient
Demographics, Hospital
Operations & Healthcare Costs



Vandana Padhi



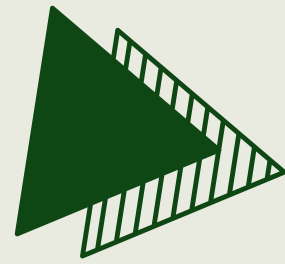
MED EX ANALYZER

The healthcare sector generates vast amounts of data from patient records, hospital operations, medical claims, and treatment outcomes. However, without proper analysis, these insights remain hidden and decision-making becomes difficult.

MedEx Analyzer aims to convert raw medical and financial data into meaningful insights that support better decision-making and improved healthcare outcomes.

Why This Matters

- Healthcare costs are rising every year.
- Hospitals need data-driven insights to optimize operations and patient management.
- Insurance providers require clear visibility into claim patterns and risk categories.



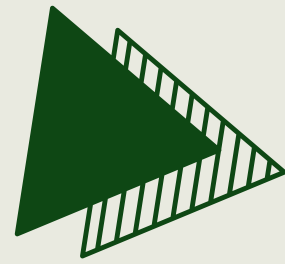
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IMPORTANCE OF THE ANALYSIS

- Better financial planning for hospitals and healthcare providers
- Improved patient care strategies through demographic insights
- Identification of high-risk patients based on BMI, age, diseases, and smoking habits
- Detection of cost drivers such as hospital type, category of charges, and region
- Data-driven decision-making across the healthcare ecosystem

REAL-WORLD PROBLEMS ADDRESSED

- Rising medical charges with no clear cost breakdown
- Hard to identify high-risk patients driving higher expenses
- Limited visibility into hospital-wise claim patterns
- Difficulty analyzing patient demographics affecting costs
- No unified view of financial, hospital, and patient insights
- Slow decisions due to manual, non-visual reporting



MED EX ANALYZER

TARGETED USERS

Primary Users

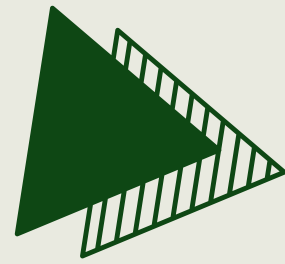
- Hospital Management & Administration
- Doctors & Medical Professionals
- Finance & Billing Department
- Healthcare Analysts & Data Teams
- Insurance & Claims Teams

Secondary Users

- Operations Teams
- Policy Makers / Healthcare Executives
- Quality & Compliance Teams
- Public Health Researchers

WHAT THE ANALYSIS FOCUSES ON

- Medical charges & cost distribution
- Patient demographics & health risk segments
- Hospital performance & operational efficiency
- Disease-wise & category-wise medical charges
- Yearly trends in claims & expenditure



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DASHBOARD OVERVIEW

Executive Overview

Provides high-level KPIs and yearly trends, along with a quick category summary for overall performance understanding.



Patient Demographics

Analyzes age, gender, BMI, diseases, and smoker distribution to understand patient profiles and risk groups.



Hospital Performance

Compares claims and charges across hospitals and highlights public vs private performance differences.



Financial & Claims Analysis

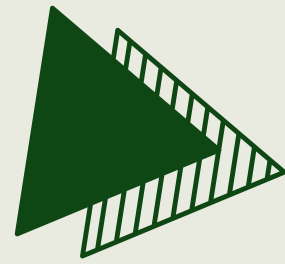
Shows category-wise charges and claims with clear year growth trends for financial monitoring.



Health Risk Insights

Identifies high-cost patients and analyzes BMI, smoking, and disease impact on medical expenditure.





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DATASET DETAILS



Dataset Source (Sample Dataset)

- Three sample CSV files used:
- patients.csv, hospitals.csv, medical_claims.csv
- Imported into **SQL Server** using Table Import Wizard
- Loaded into Power BI for cleaning, modeling & visualization

Patients Table

- Rows: 2,000 & Columns: 7
- Fields: patient_id, age, sex, bmi, smoker, location, chronic_disease
- Purpose: Contains patient demographics & health info

Hospitals Table

- Rows: 7 & Columns: 4
- Fields: hospital_id, hospital_name, location, hospital_type
- Purpose: Hospital category & location details

Medical Claims Table

- Rows: 18,000 & Columns: 6
- Fields: claim_id, patient_id, hospital_id, year, month, charges
- Purpose: Financial transactions & yearly claim trends

TOOL USED

POWER BI

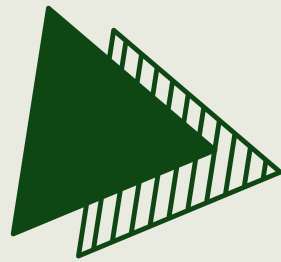
POWER QUERY

SQL

CSV

DAX

MICROSOFT FABRIC



MED EX ANALYZER

Phase 1: Data Loading

- The sample dataset was provided in 3 CSV files (patients, hospitals, claims).
- These were imported into SQL Server using the Table Import Wizard to ensure structure, integrity, and proper keys.
- SQL tables were then loaded into Power BI using Import Mode for faster performance and full DAX support.

Phase 2: Data Cleaning

- Cleaned null values, corrected data types, removed duplicates, and standardized fields.
- Created calculated columns like age_group, bmi_category, quarter, and charge_category.
- Prepared clean and analysis-ready tables for modeling.

Phase 3: Data Modeling

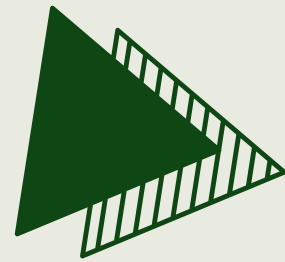
- medical_claims used as the Fact table; patients and hospitals used as Dimension tables.
- One-to-Many relationships created using patient_id and hospital_id with single-direction filters.
- Star schema chosen for simplicity, performance, and easy filtering across dashboards.

Phase 4: Creation of Measures

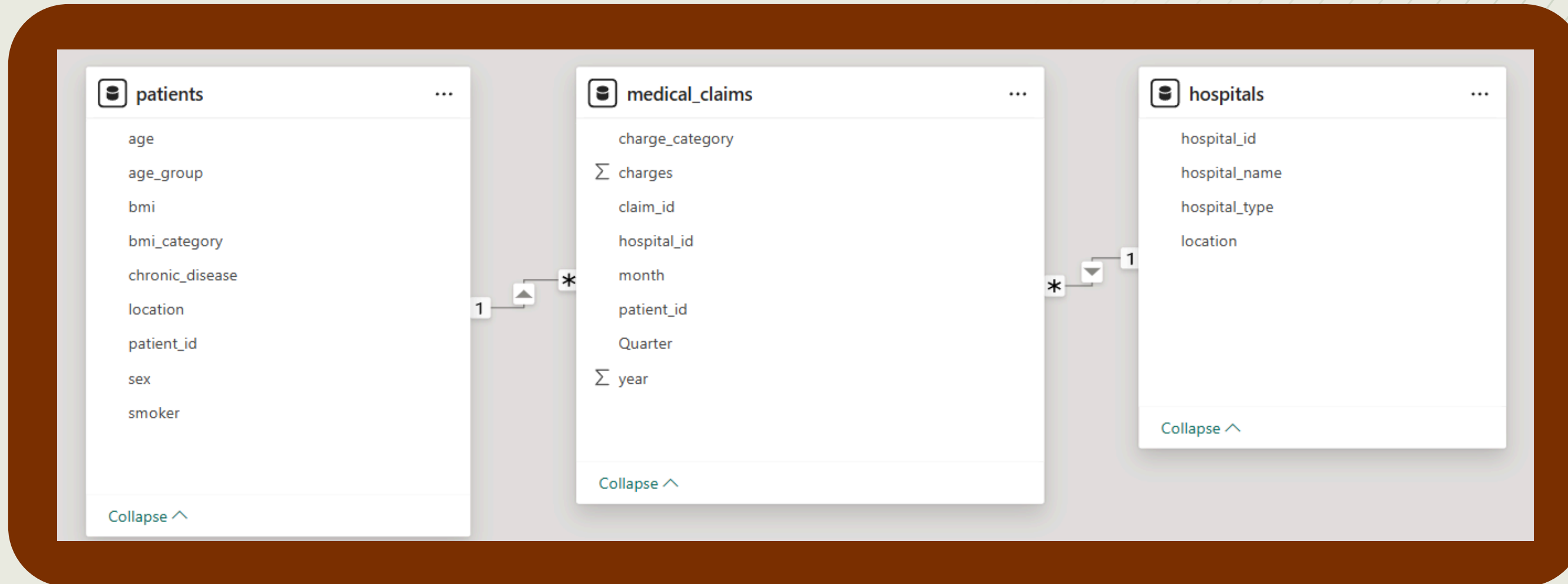
- Built core metrics such as Total Charges, Total Claims, Avg Charges, YOY Growth, and category-based calculations.
- Measures ensure accurate, dynamic KPIs that respond instantly to filters and slicers.
- Essential for delivering meaningful, interactive insights in the dashboard.

Phase 5: Dashboard Creation

- Created five interactive dashboards: Executive Overview, Patient Insights, Hospital Performance, Financial Analysis, and Health Risk Insights using KPIs, bar/line charts, donut charts, maps, and trend visuals.
- Designed to provide a complete 360° analysis of medical expenditure, patient risks, and hospital performance.
- Published and hosted the entire Power BI project on Microsoft Fabric, enabling secure access, sharing, and automated data refresh.



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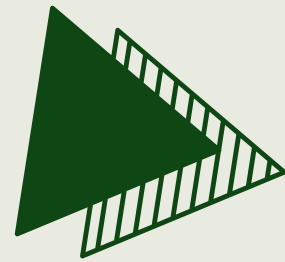


RELATIONSHIP

- Cardinality: One-to-Many
- Filter Direction: Single direction (Dimension $\rightarrow$ Fact)
- Reason: Ensures clean filtering without circular dependencies.

FACT TABLE ROLE

The **medical_claims** table sits at the center of the model because it contains measurable data (charges) and connects to both dimensions.

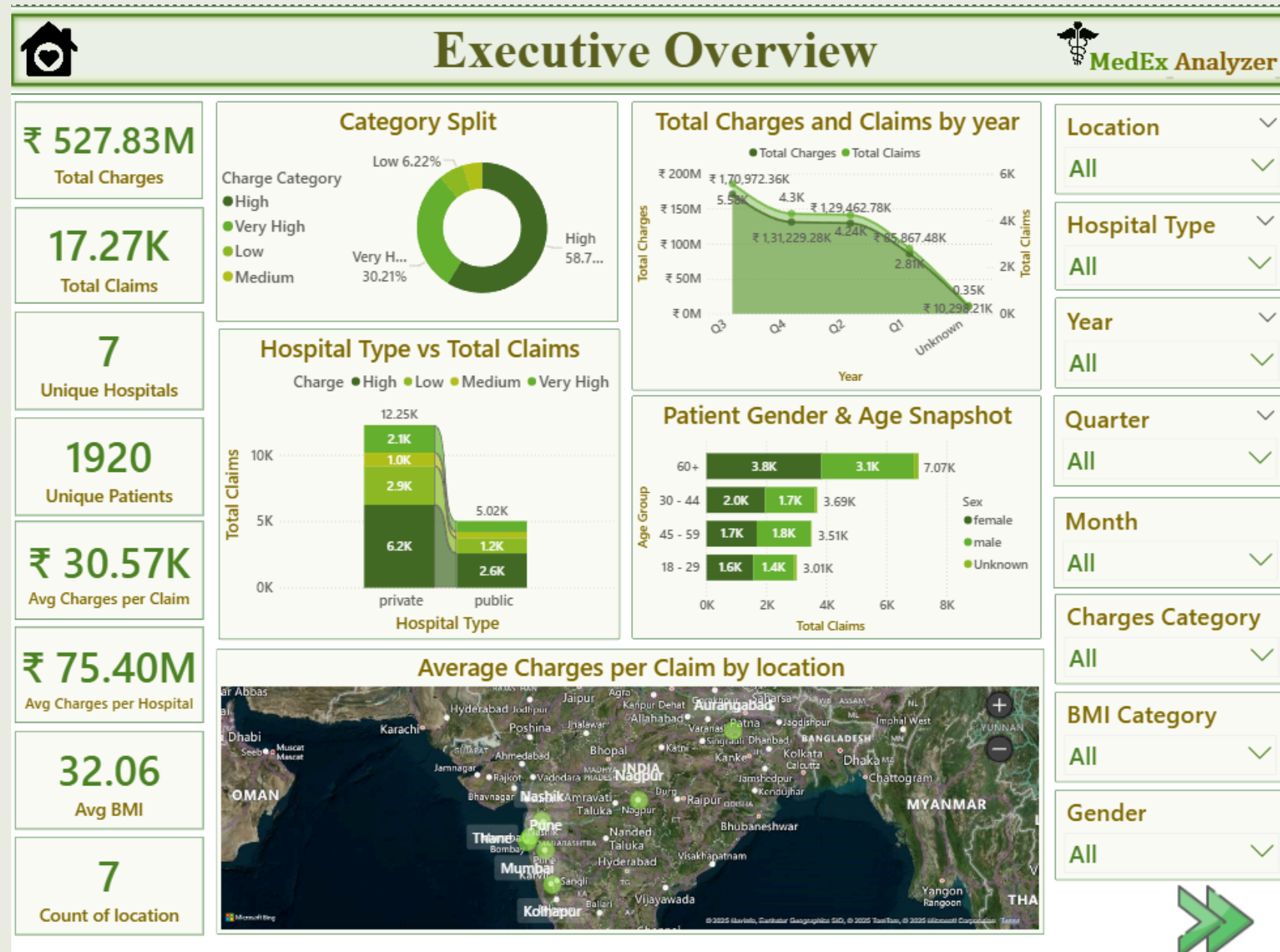


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Purpose

Gives a quick high-level summary of overall medical expenditure, patient volume, and hospital activity. Helps management understand yearly trends and major contributors at a glance.

EXECUTIVE OVERVIEW DASHBOARD

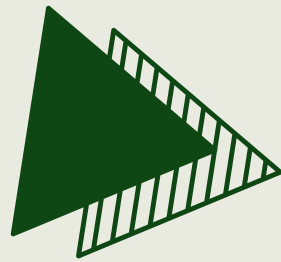


Key Visuals

- KPIs: Total Charges, Total Claims, Avg Charges per Claim, Unique Patients, Avg BMI
- Donut Chart: Charge Category Split
- Area Chart: Total Charges & Claims by Year
- Column Chart: Hospital Type vs Total Claims
- Bar Chart: Patient Gender & Age Snapshot
- Map: Average Charges per Claim by Location

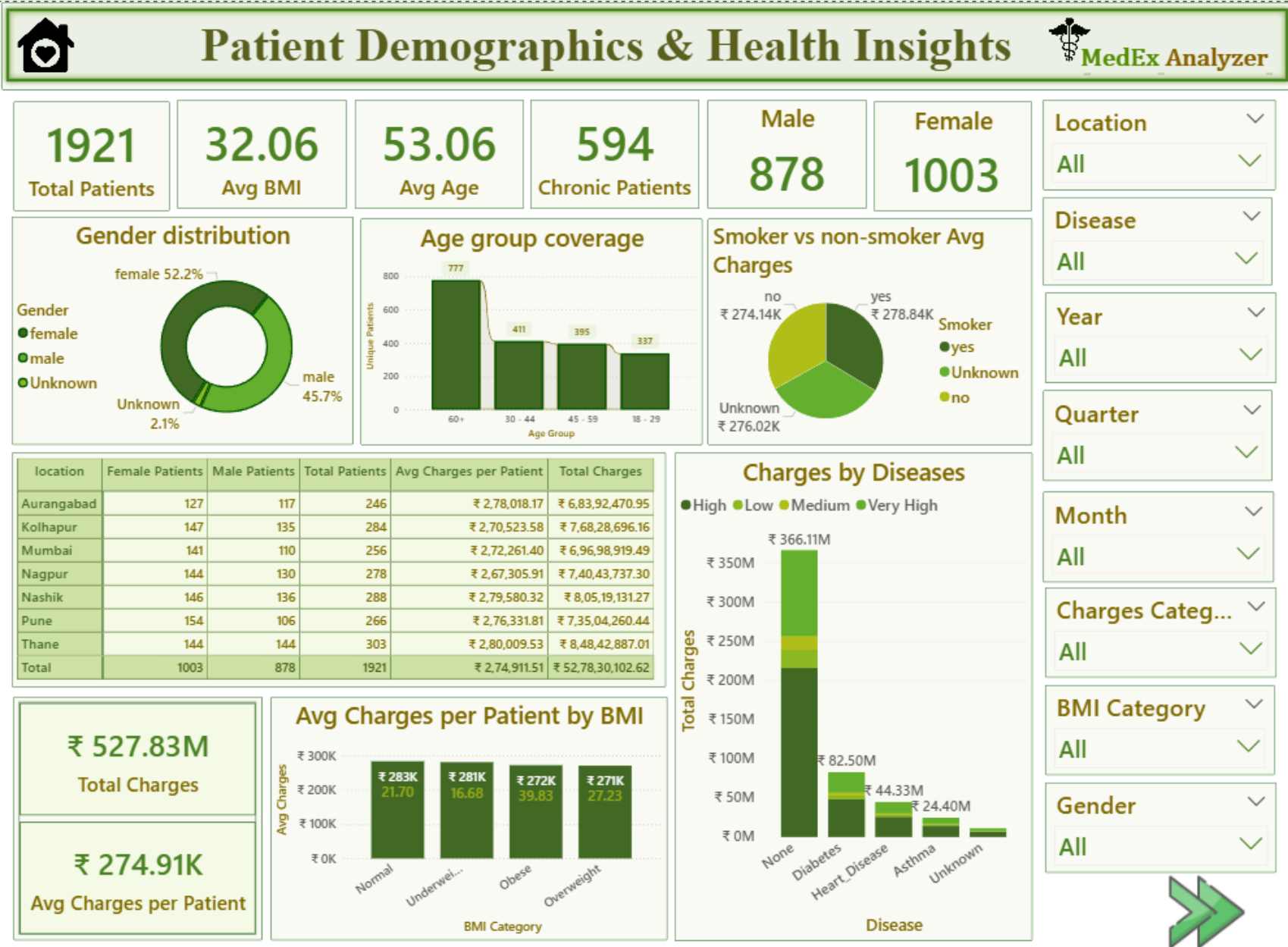
Key Insights

- High charges dominate the expenditure share.
- Private hospitals handle significantly more claims than public hospitals.
- Age groups 45–59 and 60+ show strong claim volumes.
- Certain locations reflect higher average charges on the map.



MED EX ANALYZER

PATIENT DEMOGRAPHICS & HEALTH INSIGHTS



Purpose

Provides a detailed view of patient distribution, health conditions, and risk indicators. Helps understand which age groups, BMI categories, genders, and diseases contribute most to overall charges.

Key Visuals

- KPIs: Total Patients, Avg BMI, Avg Age, Chronic Patients, Male/Female Count
- Donut Chart: Gender Distribution
- Column Chart: Age Group Coverage
- Pie Chart: Smoker vs Non-Smoker Avg Charges
- Column Chart: Charges by Diseases
- Table: City-wise patient stats & Avg Charges
- Column Chart: Avg Charges per Patient by BMI

Key Insights

- Females represent a slightly higher share of total patients.
- Age groups 60+ and 45-59 record the highest patient counts.
- Smokers and non-smokers show similar average charges with slight variation.
- Chronic diseases like Diabetes and Heart Disease drive significantly higher total medical charges.
- Obese BMI category shows the highest average charges per patient.



- ## Key Visuals

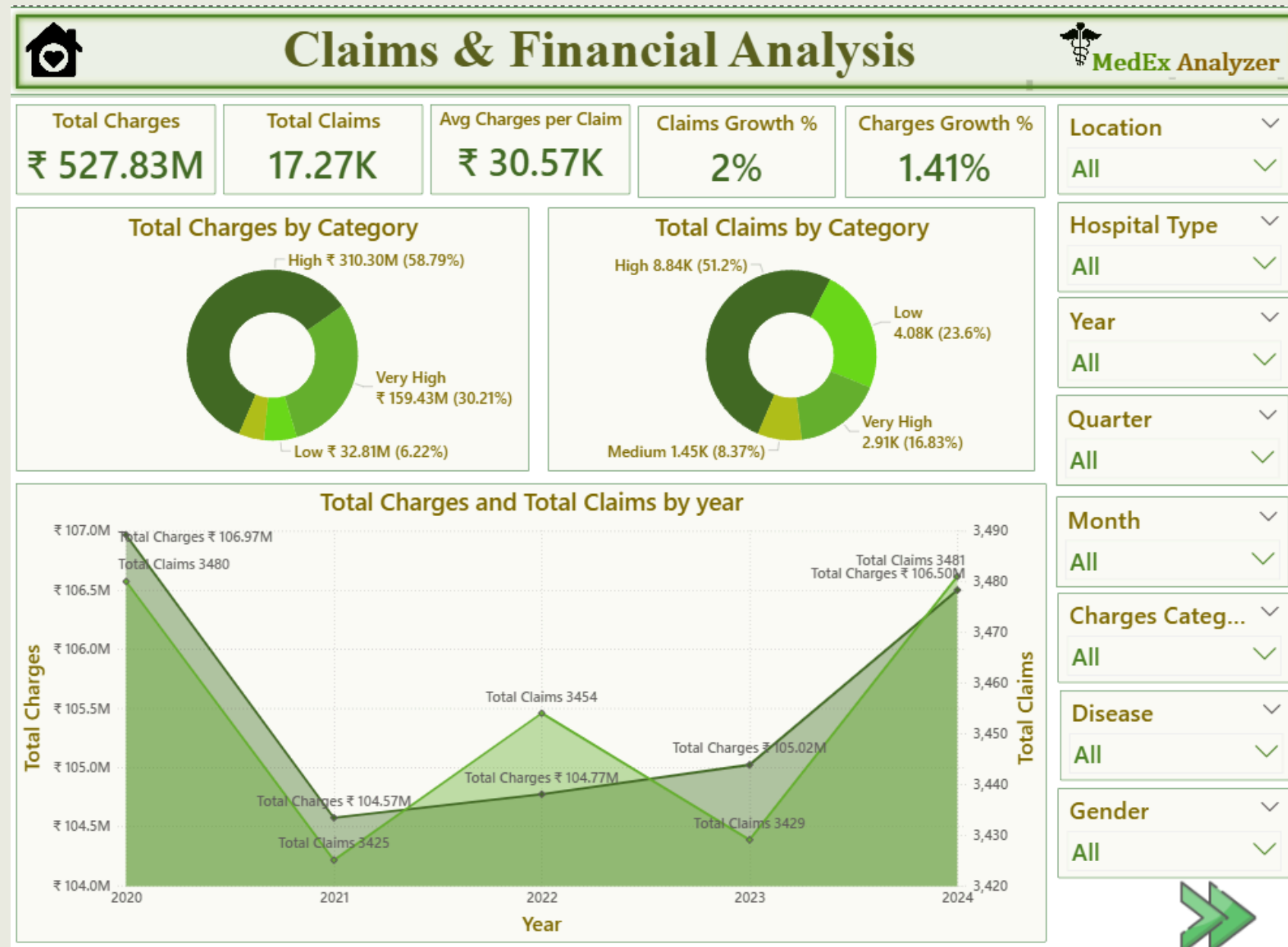


Purpose

Analyzes financial performance through category-wise charges, claims volume, and year-over-year growth.
Helps finance and management teams track cost trends and identify high-expense segments.

- KPIs: Total Charges, Total Claims, Avg Charges per Claim, Claims Growth %, Charges Growth %
- Donut Chart: Total Charges by Category (High, Medium, Low, Very High)
- Donut Chart: Total Claims by Category
- Line/Area Chart: Year-wise Total Charges vs Total Claims (2020-2024)
- Slicers: Location, Hospital Type, Year, Quarter, Month, Category, Disease, Gender

- High-charge category contributes the majority of overall medical expenses.
- Claims volume is highest in the Low and High categories, indicating diverse cost patterns.
- Charges and claims show a fluctuating pattern across years, with a clear upward trend in the latest year.
- Financial growth metrics (claims & charges) show positive percentage growth.



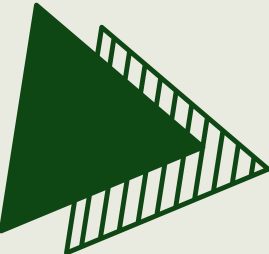


Purpose

Key Visuals

- ## Key Insights

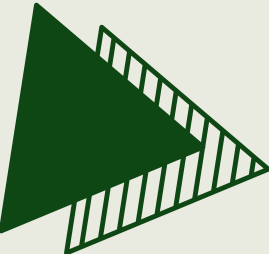
- Over 99% of patients fall under high-cost categories, highlighting expense-heavy conditions.
- Obese BMI category contributes the highest total charges, followed by Normal BMI.
- Older age groups (45-59 and 60+) incur significantly higher average charges.
- Smokers and non-smokers show similar average charges but smokers trend slightly higher overall.
- Chronic diseases like Diabetes and Heart Disease increase costs across BMI ranges.



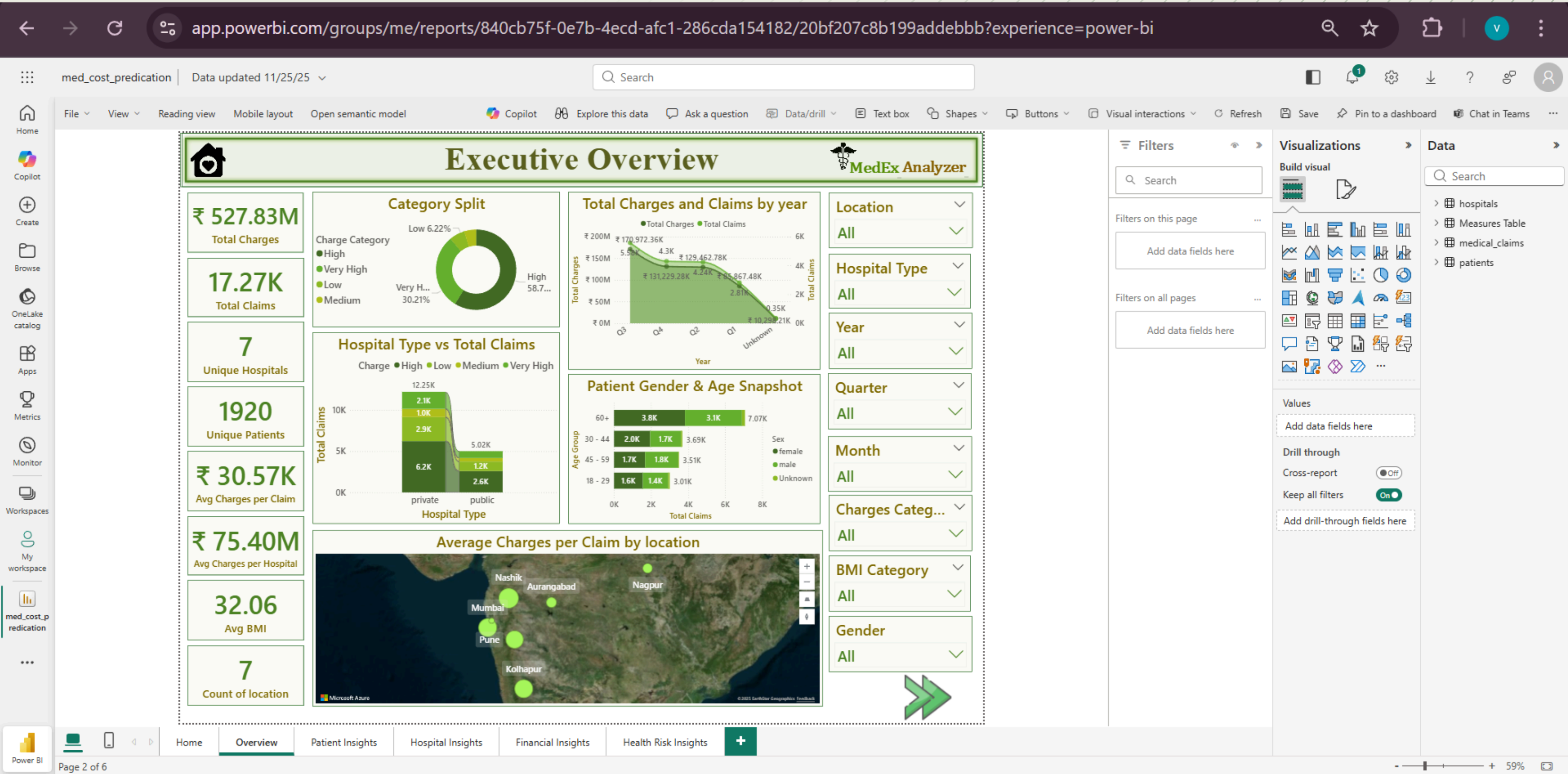
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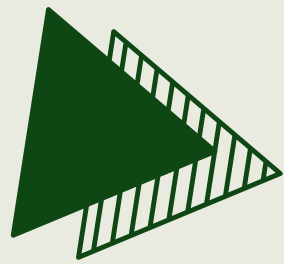
HOSTING ON MICROSOFT FABRIC

PURPOSE OF HOSTING	STEPS FOLLOWED	BENEFITS OF FABRIC HOSTING
• To deploy the Power BI report to a cloud platform for easy access and sharing. • To enable centralized viewing for all stakeholders. • To ensure the dashboard stays updated with scheduled refreshes. • To support secure, organization-wide collaboration.	• Published the final Power BI report to Microsoft Fabric Workspace. • Configured scheduled data refresh to keep reports updated automatically. • Set permissions to control who can view or edit the dashboard. • Organized the workspace for clean navigation and asset management.	• Cloud-based access from any device, anytime. • Automated refresh eliminates manual data updates. • Secure sharing with controlled user access. • Smooth collaboration for management, doctors, analysts, and finance teams. • Centralized, scalable environment for future enhancements.



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LIMITATIONS

DATA LIMITATIONS

- The data used is sample data, not actual hospital data.
- Only 7 hospitals are included, so comparisons are limited.
- Some details about patients and claims are basic and not very detailed.

TECHNICAL LIMITATIONS

- The dashboard is not connected to real-time data; it depends on static CSV files.
- Fewer columns and features limit deeper or advanced analysis.
- No forecasting was used in this project.

SCOPE LIMITATIONS

- The project focuses only on descriptive analysis, not predictions.
- The star schema is simple because the dataset has only 3 tables.
- Some insights may not represent real-world hospital operations fully.

FUTURE RECOMMENDATIONS

ADD MORE REALISTIC DATA

- Use a larger, real-world dataset with more hospitals, diseases, and patient attributes.
- Include more detailed medical records, treatment-level data, and cost breakdowns.

ENABLE REAL-TIME UPDATES

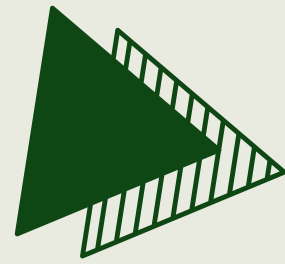
- Connect Power BI directly to an API for live data refresh.
- Automate updates using Microsoft Fabric or Dataflows.

ADD ADVANCED ANALYTICS

- Integrate forecasting, predictive modeling, or ML algorithms.
- Predict future charges, high-risk patients, or hospital load.

EXPAND SCOPE

- Include additional dimensions like medications, treatment durations, costs, or doctor performance.
- Build separate dashboards for operations, diseases, or insurance partners.



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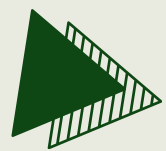
OVERALL SUMMARY

- *Built a complete healthcare analytics solution using Power BI, SQL, CSV, and Microsoft Fabric.*
- *Combined patient, hospital, and financial claim data into one unified analysis platform.*
- *Used a clean star schema and DAX measures to ensure accurate and efficient reporting.*

FINAL VALUE DELIVERED

- *Helps hospital management track performance and manage expenses.*
- *Assists doctors and analysts in understanding patient health risks.*
- *Offers a reliable dashboard for data-driven decision-making across the healthcare system.*

Thank you.



Vandana Padhi

MED EX ANALYZER

Insights into Patient Demographics, Hospital Operations & Healthcare Costs